

ChristosHadjinikolis

Engineering Manager | ML Systems Lead | UCL Research Associate
Ph.D. Computer Science | Active Committee Expert @ ISO/CEN-CENELEC JTC'21
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mediums



leadership

people management,
hiring/interviews, mentoring,
code pairing, sprint
reviews/retros, delivery
governance

practice

streaming strategy,
standardisation, semantic
contracts, **DORA metrics**,
model/data versioning,
observability

technical

production ML, signal
cleaning, streaming/lambda
backends, **Java/Kotlin, Python**,
SQL, Kafka Streams/Flink,
MLflow, Airflow, AWS/GCP

public work

practical AI standards
(JTC'21/UCL); Agile DS
talks; **dynamicio**;
skeleton-replay/IDE plugin

about me

Engineering Manager and technical ML systems leader with **16 years** of combined industry and academic experience across AI research, production ML, live data processing, streaming platforms, and backend engineering. Own Vortexa's **live streaming intelligence estate**, a major client-facing backend area where reliability and prediction trust matter daily. Over almost six years, drove the Data Production Team from **4 people to 30+**, raising operating maturity through hiring loops, system design culture, strategic platform choices, and cross-functional ways of working. Lead a **10-person cross-functional team**; UCL Research Associate and JTC'21 member advancing **practical AI standards**, auditability, and safe adoption.

work-experience

since 12/2020 **Vortexa Ltd, London, UK**

Engineering Manager / ML Tech Lead

Own Vortexa's **live streaming intelligence estate**, a major client-facing part of the company's back-end platform, filtering roughly **6M vessel-position records/hour** into production intelligence for **13.5K monitored vessels**. Lead delivery and **live-processing strategy** with a **10-person cross-functional team** spanning **6 MLE/DE/DS, 2 Product, and 2 SMEs**. Quality issues become visible quickly to clients, so reliability, explainability, and operational trust are part of the role.

- **People & Org Scaling**: Directly manage the MLE/DE/DS core and lead a **10-person cross-functional delivery team**; drove Data Production Team growth from **4 to 30+** by developing hiring loops and system design evaluations, hired **5 senior DS/DEs**, evaluated **31 candidates**, retained/progressed the full team, and onboarded new joiners to PRs within a week.
- **Streaming Transformation**: Established **Flink** as a strategic platform direction and led the two-year **Kafka Streams-to-Flink transformation** for critical live pipelines, reducing partition-coupled scaling and maintenance effort through Flink's dataflow model, independent state/checkpointing, richer operational UI, and framework-driven development.
- **Trustworthy ML Estate**: Lead the prediction trust loop across signal quality, destination/ETA, and diversion/routing outputs: **versioned models/data**, replay/evaluation interfaces, online monitoring, analyst feedback, controlled switch-over, and explainability work.
- **Reliability & Ops Culture**: Harden production with monitoring, runbooks, rollback/fallback plans, and Jira alert workflows; moved from firefighter patterns to shared on-call ownership while keeping **MTTR under 30 min**.
- **Product Semantics & Interfaces**: Facilitated cross-functional semantic alignment around recurring prediction-quality disputes, turning stakeholder friction into measurable evaluation targets and RFC/workstream-ready product interfaces.
- **Engineering Substrate**: Scale delivery through Flink/Airflow archetypes, ADRs, docs close to code, tests-as-docs, dev containers, Backstage adoption, and DynamicIO data contracts/SLAs; cut a **3h DAG feedback loop to under 5 min**.
- **Architecture & Org Influence**: Instigated a weekly company architecture forum and advocated the cross-functional topology now used around live intelligence work: engineers paired closely with Product, analysts, SMEs, and frontend ownership boundaries.

2016–2020 **DataReply, London, UK**

Data Scientist | ML Engineer

Mostly hands-on consultant across UBS, Vodafone, and CNHi; leadership was project-specific: lead DS/Scrum Master at CNHi, DS-platform contributor at Vodafone, and sole embedded consultant at UBS. Focused on productionising data science.

- **Vodafone Group**: Worked on Infinity, a GCP/**Kubeflow** DS platform, bringing hands-on DS workflow feedback into ML lifecycle support; built Red Agent, a mobile-network feature-engineering framework.
- **CNHi**: Lead Data Scientist/Scrum Master on predictive maintenance; managed/guided a DS team using **PySpark** over live vehicle sensory data with product-facing backlog discipline.
- **UBS**: Hands-on graph/process-mining analytics and observability work, building real-time pipelines with **Kafka**, **Elasticsearch**, and Python; later sole embedded DataReply consultant.

2010–2016 **Roles in UK Institutions**

Assoc. Lect. @GSM | Coding Teacher @DGC | TA @KCL|UCL

Taught Java, Python, data structures, systems architecture, semantic web, SQL, and AI at GSM, DGC, KCL, and UCL; developed clear technical communication habits during doctoral work.

education

- 2010–2014 **Ph.D. in Computer Science** [King's College London, UK](#)
***Thesis:** Persuasion Dialogues & Opponent Modelling.*
Developed logical inference methods for large **Knowledge Graphs**, combining symbolic reasoning, **graph analytics**, and Bayesian techniques (e.g., MCMC). Research contributed to scalable AI systems grounded in formal semantics. Received **Best Poster Award** at IJCAI 2013 (413 submissions). Awarded **EPSRC PhD Scholarship**, **Outstanding TA Award** (student-nominated), and earned the **Graduate Certificate in Academic Practice**.
[Access Publications on Google Scholar](#)
- 2004–2010 **Diploma (BEng) in Computer Engineering** [University of Thessaly, GR](#)
Five-year polytechnic degree in Computer Science and Engineering with a **strong focus on mathematics** (10 courses), covering theoretical aspects, networking, and telecommunications. ***Majored in Artificial Intelligence.***
- 2002–2004 **Military Officer: Second Lieutenant** [National Guard, CY](#)
Completed national service as a commissioned Second Lieutenant.

committees/affiliations

- since 10/2024 **University College London, Dept. of Information Studies** [Associate Researcher](#)
Research/teach practical AI standardisation: AI Act, auditability, model/data versioning, explainability, and safe adoption.
- since 01/2021 **ISO - International Organisation for Standardisation** [Committee Expert Member for JTC'21](#)
Member of JTC'21 Working Group 3, developing AI standards aligned with EU policy and international norms.

talks/interviews

- 2023 **Bill Raymonds, Agile in Action** [Podcast Interview](#)
Agile Data Science: Sailing Through Data Science: The Agile Journey at Vortexa, [Details: YouTube](#)
- 2022 **Open Data Science Conference** [Industry Conference Talk](#)
Dynamicio, a python library for abstracting your I/O, [Details: here](#)
- 2020 **iunera: Big Data Innovators** [Interview Blog Post](#)
The Agile approach in data science explained by an ML expert, [Details: here](#)
- 2020 **Big Data Warsaw** [Industry Conference Talk](#)
Monitoring & Analysing Communication and Trade Events as Graphs, [Details: here](#)
- 2018 **Connected Data London** [Panelist @ Industry Conference](#)
AI tribes and interconnections: what's graph got to do with it?, [Details: here](#)
- 2018 **Minds Mastering Machines** [Industry Conference Talk](#)
A talk on doing Data Science the Agile Way, [Details: here](#)

certificates & awards

- 07/2020 **AWS Certified Machine Learning - Specialty** [AWS](#)
Validation Number: D0R0WC7CN1VQ13SV
- 01/2019 **Process Mining** [Coursera Course Certificates](#)
License number: AFU2VTP6273V
- 10/2017 **Google Certified Professional - Data Engineer** [Google Cloud Certified](#)
Credential ID: 000000400
- 07/2017 **Advanced Elasticsearch: Data modelling** [Elastic](#)
Enrolment Id: 33442
- 06/2016 **Graph Analytics for Big Data** [Coursera Course Certificates](#)
License number: NZADYAQ7N8FW
- 05/2016 **Apache Spark Developer Certification** [O'Reilly Media](#)
License number: 1.x-0548
- 04/2016 **Neo4j Certified Professional** [Neo Technology, Creators of Neo4j](#)
License number: here